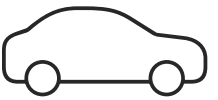


Sender V_i
(Broadcaster)

From Phase 2 proof cache (precomputed, message-independent):

Complete Groth16 proof slots: each pre-committed to predicted BSM
(position, velocity, heading, timestamp)



Receiver V_j
(Any nearby vehicle)

1

Prediction accuracy check

Receive safety message m (e.g., BSM)

$h_m = H_{\text{Poseidon}}(m \parallel t_{\text{current}})$ check: $h_m \stackrel{?}{=} h_m^{\text{pred}}$ in dequeued slot

2

Proof slot broadcast

Dequeue next slot $\pi^{(s)}$ from proof cache

Proven statement:

1. $\text{VerifyMT}(\pi_i^{\text{MT}}, \text{leaf}_i, R) = 1$
2. $t_{\text{start}} \leq t_{\text{current}} \leq t_{\text{end}}$
3. $h_m = H_{\text{Poseidon}}(m \parallel t_{\text{current}})$

Public inputs: $R, t_{\text{current}}, h_m$

Private witness: AST_i fields, π_i^{MT}

*Online cost: 0.410 ms Poseidon hash only
(no EC operations online, proof already complete)*

Latency key insight

Phase 2 precomputes:

- Complete Groth16 proofs offline
- Each committed to predicted BSM content

Phase 3 online cost:

- 0.410 ms Poseidon hash (prediction check only)

⇒ Sub-millisecond per-message overhead

$m, t_{\text{current}}, \pi, R$

3
Sender transmits over the V2V broadcast channel (e.g., DSRC / C-V2X):

Proof verification

Recompute: $h'_m = H_{\text{Poseidon}}(m \parallel t_{\text{current}})$

Look up current Merkle root R
(from last broadcast or pre-fetched)

Check timestamp: t_{current} is fresh

Run Groth16 verifier:

$\text{Verify}(\pi, R, t_{\text{current}}, h'_m) \stackrel{?}{=} 1$

REJECT
message discarded

fail

Valid?

pass

ACCEPT

V_j trusts m as authentic
No linkability to V_i 's identity